Neel Gali

Senior Software Engineer

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Professional Experience

Senior Software Engineer *Aug 2023 – Present*

DigitalOcean | Cloud Infrastructure Platform

- Architected and developed high-performance distributed systems using **Java 21** and **Python 3.12**, processing 500,000+ API requests per second with sub-100ms latency across greenfield cloud infrastructure initiatives.
- Led system architecture and design decisions for 3 greenfield microservices, collaborating with cross-functional teams to define technical strategy, evaluate trade-offs, and establish scalable production-grade solutions.
- Optimized performance-critical algorithms and data structures for resource scheduling systems, implementing advanced concurrency patterns in **Java** to reduce memory overhead by 40% and improve throughput.
- Mentored 5 junior engineers on system design principles, code quality, and best practices, participating in technical interviews and hiring decisions to expand the engineering team.
- Designed and implemented RESTful APIs and event-driven architectures using **Java Spring Boot** and **Python FastAPI**, ensuring backwards compatibility and API versioning for production services.
- Built distributed data processing pipelines using **Apache Kafka** and **Python** to handle real-time telemetry streams, applying algorithmic optimizations to process 10TB+ data daily.
- Contributed to technical strategy discussions and architectural reviews, evaluating scalability, fault tolerance, and performance characteristics of proposed system designs aligned with business objectives.
- Leveraged modern AI coding tools to accelerate development velocity, improve code quality through automated testing, and enhance debugging workflows for complex distributed systems.

Software Engineer *Jan 2021 – Jul 2023*

MongoDB | Database Infrastructure

- Developed high-performance database query optimization engine using **Java**, implementing advanced algorithms for query planning, index selection, and execution that improved query performance by 35%.
- Built scalable backend services in **Python** and **Java** for database monitoring and diagnostics, designing efficient data structures to support real-time analysis of cluster performance metrics.
- Participated in system architecture design for greenfield observability platform, contributing technical insights on distributed tracing, metrics aggregation, and log analysis at scale for production deployments.
- Implemented graph algorithms and tree traversal techniques in **Java** for query optimization, applying computer science fundamentals to solve complex problems in distributed database systems.
- Collaborated with engineering teams across time zones to deliver production-grade software, conducting code reviews, writing comprehensive tests, and ensuring adherence to engineering best practices.
- Developed internal tooling using **Python** to automate performance testing and benchmarking workflows, reducing manual testing time by 60% and improving release velocity.
- Contributed to hiring efforts by conducting technical interviews focused on algorithms, data structures, and system design, evaluating candidates for software engineering positions.

Software Developer *Jul 2019 – Dec 2020*

ThoughtWorks | Software Consultancy

- Developed microservices-based applications using **Java Spring Boot** and **Python Django**, implementing RESTful APIs and event-driven architectures for enterprise clients in fintech and healthcare.
- Applied algorithms and data structures knowledge to optimize backend systems, implementing efficient search algorithms, caching strategies, and database indexing to improve application performance.
- Built scalable data processing pipelines in **Python**, designing ETL workflows that processed millions of records daily while maintaining data integrity and system reliability.
- Contributed to system design discussions and technical architecture planning, evaluating technology choices and proposing solutions aligned with client requirements and engineering best practices.
- Wrote clean, maintainable code following SOLID principles and design patterns, participating in pair programming sessions and code reviews to ensure high-quality software delivery.
- Developed automated testing frameworks using **JUnit**, **pytest**, and integration testing tools, achieving 85%+ code coverage for mission-critical business logic.

Skills

Languages: Java (Java 23, Java 21, Spring Boot), Python (3.12, FastAPI, Django), SQL
System Design: Distributed systems, microservices architecture, high-performance systems, scalability, fault tolerance
CS Fundamentals: Algorithms, data structures, system design, concurrency, performance optimization
Technologies: Apache Kafka, RESTful APIs, event-driven architecture, database optimization, cloud infrastructure
Practices: Code reviews, automated testing (JUnit, pytest), CI/CD, agile methodologies, technical mentorship
Leadership: Technical strategy, architecture decisions, mentorship, hiring, cross-functional collaboration, AI integration

Education

Master''s in Computer Science, Oklahoma State University, Stillwater

Bachelor''s in Computer Science, JNTU Hyderabad, India